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Optimal sensor placement for leak location in water distribution networks: A feature selection method combined with graph signal processing

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ABSTRACT

Nowadays, real-time monitoring of smart water networks is essentially performed by Supervisory Control and Data Acquisition (SCADA) systems and through the use of sensors strategically positioned in the water distribution network (WDN). The purpose of sensor placement is to reasonably collect signals and thus improve the efficiency of subsequent leak diagnosis, while also limiting their deployment and operational costs. Most current sensor placement methods rely on well-calibrated hydraulic models for node selection and subsequent leak location evaluation. In this study, an efficient Optimal Sensor Placement (OSP) method for WDN is proposed, which avoids the reliance on the hydraulic model in the node selection step. It can be generalized as an optimization process, and the algorithm to solve the problem is a heuristic algorithm. We consider the user nodes in the WDN as features and propose a feature selection algorithm combined with graph signal processing. By this method, the importance of different nodes in the WDN is analyzed and the redundancy between different nodes is considered in the iterative selection process. A graph signal reconstruction method is applied to recover pressure data of all nodes using the data monitored by the selected nodes, and the relationship between the number of sensors and the reconstruction error is analyzed. The proposed method is tested on two benchmark networks of different sizes and types. A comparative analysis with other methods shows that the proposed OSP method can be easily extended to other WDNs. In addition, the proposed method achieves better monitoring results while selecting nodes quickly.

1. Introduction

Water distribution networks (WDNs) are important urban infrastructure, and the supporting facilities for a WDN in a small to medium-sized city can contain pipelines spanning hundreds of kilometers and connecting hundreds of water network user nodes (Alves et al., 2021). Once a leakage event occurs, it will cause serious water wastage and great disturbance to residents' daily life. How to monitor WDNs accurately and efficiently is urgently needed to be addressed. In the past, accurate identification of leak location in WDNs required the use of high-precision acoustic leak detection instruments and other equipment, such as Permalog acoustic devices (Khorshidi et al., 2020), which have problems including high costs and deployment difficulties. With the development of sensor technologies and mobile communication technologies such as LORAWAN (Petroni and Biagi, 2022), IoT (Plageras et al., 2018), and 5G (Prasad, 2015), it has become feasible to establish a wireless sensor network using low-cost pressure sensors to deal with the above problems. However, the dimensionality of the data on the large-scale topological graph is huge, and it takes a certain amount of resources to obtain the signals at each node. Therefore, it is impractical to place sensors at each node in the network to monitor the pressure. This requires information monitored by a small number of sensors to estimate the operating state of the entire network (Zeng et al., 2018) and provide data for subsequent fault detection in the WDN. In addition, different sensor layouts, even with a similar number of sensors, may generate different levels of redundancy (i.e., non-unique information) (Khorshidi et al., 2020). How to use a limited number of pressure sensors to monitor WDN has become an important research topic in recent years.

A variety of frameworks have been developed by researchers for the optimal sensor placement (OSP) problem. The OSP problem can be considered as an optimization problem (Rathi and Gupta, 2014), and the optimal sensor configuration can be obtained theoretically using an exhaustive search approach, but it is often difficult to implement. Myrna V. Casillas et al. proposed a semi-exhaustive search method based on an inert evaluation mechanism to deal with optimal placement for low-complexity scenarios. For high-complexity scenarios, the genetic

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algorithm (GA) and the particle swarm optimization algorithm (PSO) were introduced for stochastic optimization (Casillas et al., 2015). Li et al. (2018) developed a Modified Fruit Fly Optimization Algorithm (MFFOA) for fault diagnosis (Li et al., 2018). Mohammad S. Khorshidi et al. proposed a framework using information theoretic techniques that integrates value of information (VOI), trans-information entropy (TE), and cost to achieve optimal placement of sensors. This solution enhances the decision space and guarantees an objective choice of sensor positions (Khorshidi et al., 2018). Jafari et al. showed that the OSP problem is NP-hard by mathematical analysis, formulated it as a multi-objective optimization problem and then proposed an improved NSGA-III algorithm to solve the problem (Jafari et al., 2022). Soldevila et al. proposed a new method for sensor reallocation in large-scale systems considering fault isolation (Soldevila et al., 2022). The procedure was implemented through an optimization problem and worked with good results in two real networks in Spain.

In addition, the hydraulic model of WDN is widely used in the current study of the OSP problem. A method using the genetic algorithm and pressure sensitivity analysis was proposed and used for sensor placement and leakage analysis by Pérez et al. (2011). Convert the OSP problem to a binary matrix-dependent integer programming problem using a threshold value for measuring pressure differences. The method performs well under some specific conditions but is not ideal when the nodal water demand changes over time. Santos-Ruiz et al. proposed a heuristic algorithm for OSP considering the correlation of pressure between user nodes in WDN and analyzing the pressure information obtained from different nodes (Santos-Ruiz et al., 2022). Soldevila et al. considered the pressure data of each node as a feature of the WDN and completed the selection of sensor nodes using a hybrid feature selection method (Soldevila et al., 2018).

In recent years, researchers have found that the requirement for additional data can be effectively reduced or avoided by making full use of the topology of the WDN. L. Romero et al. proposed a leak location method. They estimated the state of the entire water network based on the hydraulic information collected by a set of pressure sensors and the network topology. Using a graph-based interpolation technique, the hydraulic state of the network nodes can be approximated. Comparing the leakage state with the healthy state, a set of nodes is selected as possible sources of leakage (Romero et al., 2021). Also, in a subsequent study, they proposed a model-free method for sensor placement in WDN based on the genetic algorithm optimization scheme. They pursue a distance-based minimization metric (pipe distance from the possible sensors to the complete set of nodes of the network) in the optimization process, thus eliminating the need for the hydraulic model of the WDN (Romero-Ben et al., 2022). These methods have achieved good results while reducing the computational burden. Graph theory-based approaches have also made some progress in water quality sensor placement. Ciaponi et al. proposed a strategy for applying water network partitioning (WNP) to create district metered areas (DMAs) and install water quality monitoring sensors through bi-objective optimization (Ciaponi et al., 2019). Giudicianni et al. proposed a method for monitoring water quality that without using any hydraulic information (Giudicianni et al., 2020). The method relies on a priori clustering of WDNs and installs quality sensors at the most central node of each cluster according to five topological centrality metrics, respectively. These methods provide satisfactory protection against pollution.

The sensor network monitoring the WDN is a graphical model of distributed interconnected sensors whose readings are modeled as time-dependent signals on vertices. Such signals that can be defined on topological graph nodes are called "topological graph signals", or "graph signals" for short (Ortega et al., 2018). Currently, developed signal processing techniques are mostly used to solve signal problems in regular domains such as speech recognition, image super-resolution analysis, and video compression. The complex and non-regular topology of the WDN makes it difficult to analyze the pressure graph signals collected by a large number of sensors using traditional and well-established theories. Graph signal processing (GSP), as an emerging field of signal processing research, aims to process signals at nodes in combination with the properties of the underlying topological graph (Marques et al., 2020; Ortega et al., 2018; Shuman et al., 2013), which provides a more comprehensive analytical approach to the problem of optimal sensor placement. The advantages of graph signal processing theory in solving the sensor optimization placement problem are mainly as follows:

(1) It provides a new perspective: While traditional data processing methods tend to optimize only for the data itself or for a specific application scenario, graph signal processing models the sensor location relationships as a connected and structured graph from the perspective of the graph, and uses graph-related theories and algorithms for analysis and optimization to obtain more accurate results.
(2) The correlation problem between sensors can be better handled: In the process of optimal sensor placement, the correlation between sensors usually needs to be considered. While graph signal processing can better deal with the correlation problem by constructing graphs and modeling the relationship between sensors using the graph localization operator.
(3) It is flexible and scalable: For some special cases, such as the redesign or expansion of the network, it may be necessary to re-evaluate and adjust the location of the sensors. While using the graph signal processing theory, it can adapt to different scenarios and get the corresponding arrangement scheme by simply adjusting the connection method and node properties of the graph, which has stronger flexibility and scalability.

The sensor node selection problem described above fits well with the objectives of graph signal sampling theory in graph signal processing. This paper aims to use the theory related to graph signal processing (Sakiyama et al., 2019), in combination with the idea of minimal-Redundancy-Maximal-Relevance (mRMR) in feature selection, to select the nodes for sensor placement in the WDN. Most of the existing sensor optimization methods require the use of hydraulic information from the WDN, the process involves complex data processing. And the conventional method of using hydraulic data of all nodes in the network to optimize the selection of nodes has limited application value, because in real cases, considering the actual topography and cost issues, not all nodes' hydraulic information can be monitored. So, obtaining a well-calibrated hydraulic model becomes a common problem for researchers when studying the node selection step for optimal sensor placement and the evaluation step of the leak location during the placement. Our approach requires the support of the hydraulic model only in the leak location evaluation step. And only the topological information of WDN is utilized in the node selection step, thus alleviating the reliance on the hydraulic model. The main contributions of this paper are as follows:

(1) We propose a new method for the OSP problem in WDNs, using the graph localization operator to characterize the graph structure information and information redundancy provided by different user nodes to select the optimal sensor configuration under a determined number. Compared with some existing methods, our approach is more efficient.
(2) We recover the global data from the data collected by the arranged sensors and analyze the effectiveness of the recovery when using different numbers of sensors.
(3) The proposed method reduces the dependence on the hydraulic simulation of WDN to some extent. It can be easily extended to different types of WDNs and also has good adaptability to environmental noise.

The rest of the paper is organized as follows. Section 2 introduces the theory and significance related to the graph localization operator. Section 3 describes the method of sensor placement for WDN monitoring proposed in this paper. Section 4 illustrates the method for validating the optimal sensor configuration using leak location and gives comparative evaluation metrics. Section 5 applies the proposed method to two water distribution networks of different sizes and types, tests and compares the existing method with the proposed method in this paper. Section 6 concludes the whole paper.

2. Basic theory

2.1. Basic theory of graph signal processing

The water distribution network can be described as a topology graph containing user nodes and pipelines. In this paper, we consider a WDN containing N user nodes as an undirected, connected, and weighted graph $\mathscr{G} = (\mathscr{V}, \mathscr{E}, \mathbf{A})$, where $\mathscr{V}$ is the set of vertices, which can be described as $\mathscr{V} = \{v_i\}_{i=1}^{N}$, $\mathscr{E}$ is the set of edges of all connected nodes, and $\mathbf{A} \in \mathbf{C}^{N\times N}$ is the weight matrix corresponding to the graph $\mathscr{G}$. We weight the pipeline distances by a Gaussian function, with lower values of edge weights between two points that are farther away and higher values of edge weights between two points that are closer together (Perraudin and Vandergheynst, 2017). A_{ij} in the weight matrix $\mathbf{A}$ denotes the weight of the edge that connects vertices v_i and v_j, points out the similarity or dependence of the signal values between vertices v_i and v_j (Chen et al., 2014). We use $f(i)$ to denote the value of the graph domain signal on the vertex v_i in graph $\mathscr{G}$. The complete graph signal can be written in the form of a vector, i.e. $\mathbf{f} = (f(1), f(2), \ldots, f(N))^T \in \mathbf{C}^N$.

In classical digital signal processing theory, the Fourier transform form for a signal $m(t)$ is shown in Eq. (1).

$$M(\omega) = \int_R m(t)e^{-j\omega t}dt \tag{1}$$

Here $e^{-j\omega t}$ is the characteristic function of the one-dimensional Laplacian matrix, as in Eq. (2).

$$-\Delta\left(e^{j\omega t}\right) = -\frac{\partial^2}{\partial t^2}e^{j\omega t} = \omega^2 e^{j\omega t} \tag{2}$$

The extension of Fourier analysis to graphs requires the use of graph Fourier bases, the most commonly used graph Fourier bases being the eigenvectors of the combined (or non-normalized) graph Laplacian (Perraudin et al., 2018). The graph Laplacian matrix $\mathbf{L}$ is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D} = diag\{d_i\}_{i=1}^{N}$ is called the degree matrix and d_i is the sum of the weights of the edges connected to the node v_i. Eigen-decomposition of the graph Laplacian matrix $\mathbf{L}$, yields the eigenvalue matrix

$$\Lambda = diag\{\lambda_k\}_{k=0}^{N-1}$$

where

$$0 = \lambda_0 < \lambda_1 \le \lambda_2 \le \ldots \le \lambda_{N-1} = \lambda_{\max}$$

and the eigenvector matrix $U = \{u_k\}_{k=0}^{N-1}$. The eigenvector corresponding to each eigenvalue is N-dimensional, i.e.

$$u_k = (u_k(1), u_k(2), \ldots, u_k(N))^T$$

The Fourier transform and inverse transform of the graph signal can be defined as Eqs. (3) and (4). $\widehat{f}(k)$ denotes the Fourier coefficient in the frequency domain of the graph, and $u_k(i)$ denotes the i th term of the eigenvector $\mathbf{u}_k$.

$$\widehat{f}(k) = \sum_{i=1}^{N} u_k^*(i)f(i) \tag{3}$$

$$f(i) = \sum_{k=0}^{N-1} \widehat{f}(k)u_k(i) \tag{4}$$

2.2. Graph localization operator

The graph localization operator is derived from the spectral graph wavelet transform and is generated by a wavelet operator based on the value function of the graph Laplacian operator (Hammond et al., 2011). For the spectral graph wavelet kernel g, the wavelet operator $g(\Lambda)$ acts on the given function $\mathbf{f}$ by modulating each Fourier mode to obtain $g(\Lambda)\widehat{\mathbf{f}}$. The Fourier inverse transform is used to obtain Eq. (5), where λ_k denotes the eigenvalue corresponding to u_k.

$$T_g f(\varsigma) = \sum_{k=0}^{N-1} g(\lambda_k)\widehat{f}(k)u_k(\varsigma) \tag{5}$$

Eq. (6) is then obtained by applying these operators to the impulse on a single vertex v_i.

$$\psi_{v_i} = T_g\delta_{v_i} \tag{6}$$

Explicit expansion in the graph domain yields

$$\psi_{v_i}(\varsigma) = \sum_{k=0}^{N-1} g(\lambda_k)u_k^*(i)u_k(\varsigma) \tag{7}$$

Without considering scaling, the wavelet coefficients for a given function $\mathbf{f}$ are generated by taking the inner product of these wavelets, as in Eq. (8).

$$\mathbf{W}_{\mathbf{f}}(v_i) = \langle \Psi_g(v_i), \mathbf{f} \rangle \tag{8}$$

$\Psi_g(v_i)$ as shown in Eq. (9).

$$\Psi_g(v_i) = [\psi_{v_i}(1), \psi_{v_i}(2), \ldots \psi_{v_i}(\varsigma), \ldots, \psi_{v_i}(N)]^T \tag{9}$$

If the smoothing kernel g is localized at a central vertex v_i in the graph, Eq. (9) is the graph localization operator for the vertex. The complete graph localization operator for the vertex is in the form of a multidimensional column vector. Note that it is a vertex-domain operator and takes into account the graph spectral domain information. This generalized localization operator applies to the kernel defined in the graph spectral domain. In Hammond et al. (2011), David K. Hammond et al. showed that the graph localization operator is localized in the small-scale limit. The amplitude of this operator decays as the distance between nodes and other nodes increases (Perraudin et al., 2018). So this operator can reflect the localization characteristics of the central node.

3. Methodology

3.1. Problem formulation

From the view of problem formulation, the purpose of optimal sensor placement is to select a given number s of nodes in a WDN containing many user nodes and to arrange sensors on the selected nodes so that the monitoring effect of these sensors on the WDN is maximized.

Definition 3.1 (Binary selection matrix). To better describe this problem, we define the binary vector of Eq. (10) to represent the selection of sensor nodes. Where N is the number of user nodes in the WDN and j indicates that the current vector represents the user node v_j. When a node arranges a sensor, its corresponding binary vector component is set to 1.

$$C = (c_1, c_2, \dots, c_j, \dots, c_N) \tag{10}$$

Proposition 3.1. The problem of optimal sensor placement is rather the problem of selecting the set of nodes to obtain better monitoring effects. Consider using the function $\alpha(C)$ to represent the monitoring effect of the arranged sensor nodes, and the selected sensor set C is the variable of the function α. The number of sensors to be selected is s. Then the optimal sensor placement problem can be mathematically formulated as the following optimization problem.

$$\begin{aligned} &\max_{C} \alpha(C) \\ &s.t. C = (c_1, c_2, \dots, c_j, \dots, c_N) \\ &c_j \in \{0, 1\}, \forall j \\ &\sum_{j=1}^{N} c_j = s \end{aligned} \tag{11}$$

3.2. The proposed method

The simplest method for selecting nodes is the exhaustive search method, where all possible results are tested to select the best solution. This is only achievable for small sets of nodes. Once the set of candidate nodes becomes large, this method becomes computationally difficult to handle. In our proposed method, we consider each user node in a water distribution network as a feature of this network and use the graph localization operator introduced in Section 2 to evaluate the importance of this feature. According to the analysis in Section 2.2, the graph localization operator can reflect the localization feature of the central node. In other words, the larger the value of the graph localization operator of a node, the more important the node is in the graph. Based on this, the method of sensor nodes selection for WDN is proposed.

Definition 3.2 (Graph Localization Information (GLI)). Consider the 1-norm of the graph localization operator of a vertex v_i in the graph topology as the Graph Localization Information of the node, which corresponds to Eq. (12).

$$\Phi_{GLI}(v_i) = \| \Psi_g(v_i) \|_1 = \sum_{\varsigma=1}^{N} |\psi_{v_i}(\varsigma)| \tag{12}$$

The larger the value of Φ_{GLI}, the more information the node provides about the graph topology without considering the redundancy between nodes, and the more important it is in the WDN.

To analyze the redundancy between the localization feature information provided by different vertices, we draw on the evaluation method of the cross-correlation relationship between different signals in signal processing theory.

Definition 3.3 (Cross-correlation Graph Information Redundancy (CGIR)). Consider using the magnitude of the 1-Norm of the product of the graph localization operator of two nodes to quantify the redundancy between the localization feature information. The magnitude of the Cross-correlation Graph Information Redundancy between vertex v_i and vertex v_j is given by Eq. (13), where $\circ$ denotes the multiplication of the corresponding terms of the matrix.

$$\Phi_{CGIR}(v_i v_j) = \| \Psi_g(v_i) \circ \Psi_g(v_j) \|_1 \tag{13}$$

Based on the above definition, we propose an objective function that uses a heuristic algorithm to deal with the above optimization problem and thus iteratively selects sensor nodes, as in Eq. (14).

Table 1
Sensor nodes selection algorithm.

Input: Set of user nodes $\mathscr{V} = \{v_i\}_{i=1}^{N}$. Adjacency matrix **A**. Degree matrix **D**. Binary selection matrix $C_{selected} = (c_1, c_2, \dots, c_j, \dots, c_N)$. The set of selected nodes S. Number of sensors to be selected s. The kernel function $g(\lambda)$.
Initialization: $C \leftarrow 0$, $\alpha \leftarrow 0$, $S \leftarrow \varnothing$, $t \leftarrow 0$;
$\mathbf{L} = \mathbf{D} - \mathbf{A}$
Eigen decomposition: Eigenvalue $\Lambda = diag\{\lambda_k\}_{k=0}^{N-1}$ and Eigenvector Matrix $U = \{u_k\}_{k=0}^{N-1}$
for $i = 1, \dots, n$
for $\varsigma = 1, \dots, n$
$\psi_{v_i}(\varsigma) = \sum_{k=0}^{N-1} g(\lambda_k) u_k^*(i) u_k(\varsigma)$
end for
$\Psi_g(v_i) = [\psi_{v_i}(1), \psi_{v_i}(2), \dots \psi_{v_i}(\varsigma), \dots, \psi_{v_i}(N)]^T$
end for
for sensornum $= 1, \dots, s$
for all v_i in $\mathscr{V}$
$t = \dfrac{\overline{\Psi} \| \Psi_g(v_i) \|_1}{\varepsilon + \frac{1}{|C_{selected}|} \sum_{j \in C_{selected}} \| \Psi_g(v_i) \circ \Psi_g(v_j) \|_1}$
if $t > \alpha$ **then**
$\alpha = t, v_{Nodenum} = v_i$
end if
end for
Set, $c_{Nodenum} = 1$
Set, $S = S \cup \{v_{Nodenum}\}$
Set, $\mathscr{V} = \mathscr{V} - \{v_{Nodenum}\}$
end for
Output: The set of selected nodes S.

$$\begin{aligned} &\max_{v_i} \frac{\overline{\Psi} \Phi_{GLI}(v_i)}{\varepsilon + \frac{1}{|C_{selected}|} \sum_{j \in C_{selected}} \Phi_{CGIR}(v_i v_j)} \\ &s.t. C_{selected} = (c_1, c_2, \dots, c_j, \dots, c_N) \\ &c_j \in \{0, 1\}, \forall j \\ &\sum_{j=1}^{N} c_j \le s \end{aligned} \tag{14}$$

Substituting Eq. (12) and Eq. (13) above yields

$$\begin{aligned} &\max_{v_i} \frac{\overline{\Psi} \| \Psi_g(v_i) \|_1}{\varepsilon + \frac{1}{|C_{selected}|} \sum_{j \in C_{selected}} \| \Psi_g(v_i) \circ \Psi_g(v_j) \|_1} \\ &s.t. C_{selected} = (c_1, c_2, \dots, c_j, \dots, c_N) \\ &c_j \in \{0, 1\}, \forall j \\ &\sum_{j=1}^{N} c_j \le s \end{aligned} \tag{15}$$

In Eq. (15), $C_{selected}$ is the set of currently selected nodes. $\overline{\Psi}$ is the average of the localization operator of the nodes in the WDN, whose main role is to keep the numerator and denominator in the objective function in the same order of magnitude together with the coefficient $1/|C_{selected}|$, so that the comparison between the graph localization information Φ_{GLI} and cross-correlation graph information redundancy Φ_{CGIR} is more accurate. ε is a very small value, used to exclude the very special case where the denominator is zero. When iteratively selecting nodes, the previous optimal locations are fixed and the nodes that have already been selected are removed from the set of candidate nodes. The selection is then made among the remaining nodes. With the objective function (15), we evaluate the valid information provided by the nodes and the information redundancy between the current node and the already selected nodes. Iteratively select the current best (maximum ratio) sensor node as the placement location for the pressure sensor until a given number of sensors s is reached, i.e., $\sum_{j=1}^{N} c_j = s$.

Table 1 lists the detailed pseudo-code for the optimal sensor placement using the proposed feature selection algorithm combined with graph signal processing.

3.3. Pressure signal recovery and number of sensors

In most cases, the number of sensors used for leak location is determined by the number of devices available. If sufficient resources are available to monitor the network intensively, it is necessary to take into account that increasing the number of sensors does not always produce better performance in locating leaks. In this section we give our suggested number of sensors by analyzing the reconstruction error of the pressure signal, and it will be an interesting topic for the future to determine exactly the optimal number of sensors.

Our sensor selection method for water distribution networks is essentially a sampling method for graph signals. In this section, we reconstruct the pressure data of the entire network using the pressure data collected by the selected sensor nodes by the method in the paper (Sakiyama et al., 2019), i.e., Eq. (16).

$$\mathbf{f}_R = \left[(\Psi_g)^r\right]_{NC}\left(\left[(\Psi_g)^r\right]_{CC}\right)^{+}\mathbf{f}_C \tag{16}$$

where Ψ_g denotes the matrix of the graph localization operator with dimension $N \times N$. r is called the polynomial approximation order. The larger the value of r, the better the reconstruction. Here, we take 20 uniformly for the sake of comparison. $[\cdot]_{NC}$ denotes the matrix consisting of the columns corresponding to the selected nodes in the graph localization operator matrix, $[\cdot]_{CC}$ denotes the matrix consisting of the rows and columns corresponding to the selected nodes in the graph localization operator matrix. $\mathbf{f}_C$ is the pressure signal collected by the selected sensor node and $\mathbf{f}_R$ is the reconstructed pressure signal of all nodes.

The reconstructed minimum mean-square error is shown in Eq. (17).

$$e = \| \mathbf{f}_R - \mathbf{f} \|_2^2 \tag{17}$$

As the number of sensors increases, the reconstruction error of the pressure signal gradually decreases, while the cost of sensor arrangement is increasing, and the two are negatively correlated. The benefit of increasing the number of sensors gradually decreases. In reality, the number of available sensors for sensor placement is always much less than half the number of user nodes in the WDN. We use the reconstruction error of the pressure signal when selecting half the number of user nodes as the target for sensor selection. As the number of sensors is further reduced from half of the total, valid information is lost. Considering that the reconstruction error is only 5% worse compared to the target result, while the cost is reduced by more than half. In this case, due to sensor cost constraints, we recommend deploying the number of sensors that can achieve 95% of the target. That is, the number of selected sensors is the minimum number that satisfies Eq. (18), where e_m denotes the reconstruction error when m sensors are selected.

$$\frac{e_1 - e_m}{e_1 - e_{N/2}} \geq 0.95 \tag{18}$$

4. Results evaluation method

4.1. Sensitivity matrix

The complete problem of OSP for WDNs consists of two main levels: select nodes for sensor placement and leak location to verify the sensor placement scheme. This section focuses on the sensitivity matrix that needs to be used when evaluating using the leak location method.

The pressure of the nodes in the WDN is highly correlated with the flow rate, so the change in the pressure value can be used to reflect the change in the flow rate in the network. In this paper, we mainly use the hydraulics simulation software EPANET (Rossman, 2000) and Water Network Tool for Resilience (WNTR) (Klise et al., 2018) to complete the node leakage simulation and pressure data acquisition, WNTR is a Python package compatible with EPANET.

First, the pressure data of all nodes in the WDN when the water demand is normal is simulated to obtain the pressure matrix

$$P_0 = \{p_{v_1}, p_{v_2}, \ldots, p_{v_i}, \ldots, p_{v_N}\}$$

under normal conditions, where p_{v_i} denotes the pressure value of node v_i during normal operation. When a leak occurs, the pressure of the node read from the leak scenario is different from the expected pressure obtained from the non-leak scenario simulation. A similar behavior can be observed when the added demand is allocated to a node (Khorshidi et al., 2020). In this paper, we perform leakage simulation by increasing the water demand node by node. The difference between the pressure at the time of leakage and the normal pressure is used to obtain the pressure residual, as in Eq. (19).

$$r_{v_iv_j}(\rho) = p_{v_iv_j}(\rho) - p_{v_i}R \tag{19}$$

where $r_{v_iv_j}(\rho)$ represents the pressure residual at node v_i when node v_j has a leak level of size ρ and $p_{v_iv_j}(\rho)$ is the pressure at node v_i when node v_j has a leak level of size ρ. The pressure residual matrix R can be expressed as

$$R = \begin{bmatrix} r_{v_1v_1}(1) & r_{v_2v_1}(1) & \cdots & r_{v_Nv_1}(1) \\ r_{v_1v_1}(2) & r_{v_2v_1}(2) & \cdots & r_{v_Nv_1}(2) \\ \vdots & \vdots & \ddots & \vdots \\ r_{v_1v_N}(M) & r_{v_2v_N}(M) & \cdots & r_{v_Nv_N}(M) \end{bmatrix} \tag{20}$$

To simulate various errors (e.g., measurement errors, statistical errors, etc.) in real situations, appropriate noise can be added to the pressure residual matrix. The sensitivity matrix can be obtained by dividing the terms in the pressure residual matrix by the corresponding leakage size as in Eq. (21),

$$S = \begin{bmatrix} \frac{r_{v_1v_1}(1)}{d_{leak}^1(1)} & \frac{r_{v_2v_1}(1)}{d_{leak}^1(1)} & \cdots & \frac{r_{v_Nv_1}(1)}{d_{leak}^1(1)} \\ \frac{r_{v_1v_1}(2)}{d_{leak}^1(2)} & \frac{r_{v_2v_1}(2)}{d_{leak}^1(2)} & \cdots & \frac{r_{v_Nv_1}(2)}{d_{leak}^1(2)} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{r_{v_1v_N}(M)}{d_{leak}^N(M)} & \frac{r_{v_2v_N}(M)}{d_{leak}^N(M)} & \cdots & \frac{r_{v_Nv_N}(M)}{d_{leak}^N(M)} \end{bmatrix} \tag{21}$$

where $d_{leak}^N(M)$ denotes the leak size corresponding to a leak of leak level M at node N.

4.2. Leak location evaluation indicators

The purpose of the sensor placement optimization problem for WDNs is to better localize leakage events. The selection of the leak location method usually affects the evaluation of the sensor placement method. Here we use a generic leak location method for evaluation. The leak localization method used is based on a classifier that uses supervised learning techniques to identify directional patterns of pressure residues, as described in Santos-Ruiz et al. (2020)). We use the K-nearest neighbor classification algorithm (KNN) in machine learning (Li and Li, 2022). The leak localization algorithm uses the pressure sensitivity matrix S introduced in Section 4.1 and the corresponding leak labels as the training feature set for the machine learning classifier. Each row of the pressure sensitivity matrix corresponds to a leak node, and the leak node of the data is the label of the data in this row. Based on the obtained sensor configuration, the pressure data collected by each sensor at the time of the leak is used to predict the leak location.

The pressure variations of adjacent nodes in the WDN are similar, which leads to an impact on the accuracy of predicting the leak location using the machine learning classifier algorithm. The predicted nodes are

Table 2
Details of WDNs used for demonstration.

Name	Number of nodes	Number of pipes	Average daily water demand (L/s)	Total pipe length (km)
Hanoi	31	34	5538.9	39.42
Net 3	92	117	3052.11	215.71

often distributed in the vicinity of the leak nodes, so it is not appropriate to use the accuracy of leak localization as an evaluation criterion. A more effective evaluation method is the average topological distance (ATD) mentioned by Li et al. (2019), which is given by

$$ATD = \frac{\sum_{i=1}^{N}\sum_{j=1}^{N}\omega_{ij}D_{ij}}{\sum_{i=1}^{N}\sum_{j=1}^{N}\omega_{ij}} \tag{22}$$

Where N is the number of nodes in the WDN, ω_{ij} denotes the number of leaks occurring at node v_i predicted for node v_j, and D_{ij} is the topological distance of the shortest pipeline path between node v_i and node v_j. The ATD can well reflect the actual distance between the predicted node and the leak location. In our method, the accuracy of leak location is measured using the ratio of ATD to the total length L of the pipeline network, which is called the average topological distance index ξ_{ATD}, and is given by

$$\xi_{ATD} = \frac{\sum_{i=1}^{N}\sum_{j=1}^{N}\omega_{ij}d_{ij}}{L\sum_{i=1}^{N}\sum_{j=1}^{N}\omega_{ij}} \tag{23}$$

The root mean square error (RMSE) is often used to assess the stability of the prediction results. Assuming that there are N samples and $\widehat{y}_i$ is the predicted value of y_i, the general formula for the RMSE is

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y}_i\right)^2} \tag{24}$$

According to the analysis by Li et al. (2019), combining ATD with RMSE can evaluate the deviation of the average topological distance between the predicted and the actual nodes, and the root mean square error of ATD is given by

$$RMSE_{ATD} = \sqrt{\frac{\sum_{i=1}^{N}\sum_{j=1}^{N}\omega_{ij}D_{ij}^2}{\sum_{i=1}^{N}\sum_{j=1}^{N}\omega_{ij}}} \tag{25}$$

The ratio of $RMSE_{ATD}$ to the total length L of the pipe network is used to measure the deviation of the leak localization and its stability, called the average topological distance deviation index μ_{RMSE}, and is given by

$$\mu_{RMSE} = \frac{RMSE_{ATD}}{L} \tag{26}$$

5. Results and discussion

According to whether the network contains devices such as pumps and valves, WDNs can be divided into two broad categories: active WDN and passive WDN. In this section, we apply the proposed sensor placement method to two different types of WDNs, the passive Hanoi Water Network in Vietnam and the active North Marin Water District (Net 3) in Novato, California, USA. Both two models have been widely used in the study of OSP in WDNs and can be downloaded from http://www.uky.edu/WDST/database.html. Some parameters of the networks are given in Table 2. In addition, our method is compared with several other methods. All work in this paper was performed on a laptop configured with Intel(R) Core(TM) i7–9700F CPU @ 3.00 [GHz], 16 [GB] RAM, NVIDIA GeForce RTX 2070, and MATLAB version is MATLAB 2020b.

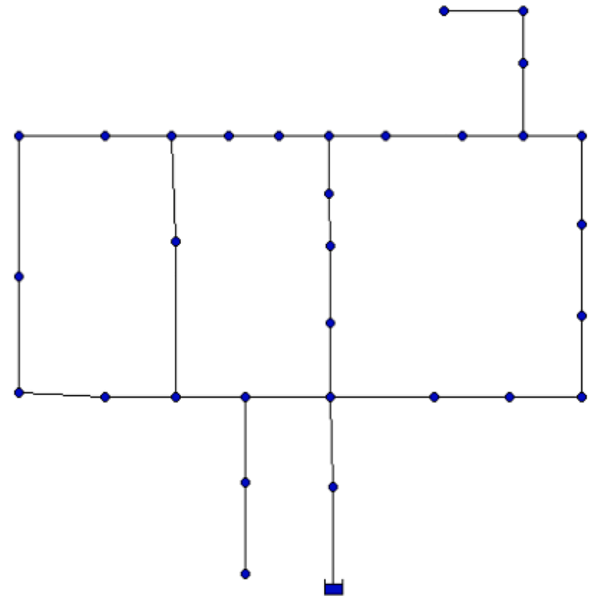

Fig. 1. Hanoi network.

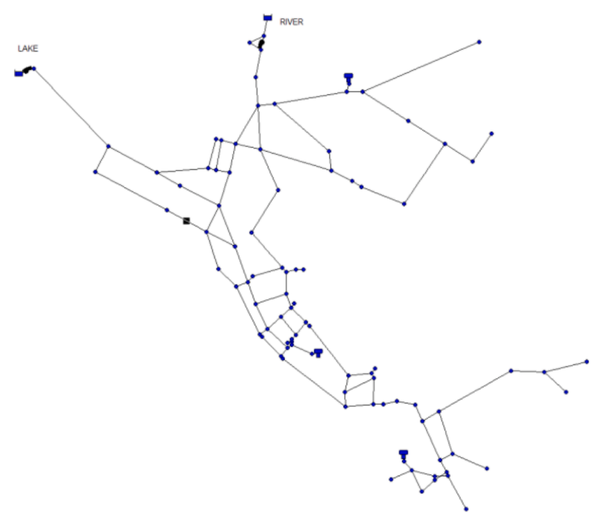

Fig. 2. North Marin Water District (Net 3).

5.1. Dataset generation

The WDN used in Case 1 is the Hanoi network with the topology shown in Fig. 1, which was proposed by Fujiwara and Khang (1990) based on the trunk network in the WDN plan of Hanoi, the capital of Vietnam (Fujiwara and Khang, 1990). The Hanoi system contains 2 reservoirs and 31 customer nodes connected by 34 pipes with pipe sizes ranging from 12 ~ 40 inches and a total length of 39.42 km. The average daily water demand is 5538.9 L/s.

The data required for leakage location in the WDN is simulated using EPANET software, and leaks occurring at nodes in the network can be achieved by adding water demand. First, we ran the Hanoi system network in EPANET software to obtain the pressure data under normal conditions, and then add different sizes of water demand node by node to simulate leaks. Note that the size of the leak should not be too large, otherwise, it will affect the normal operation of the network, and the size

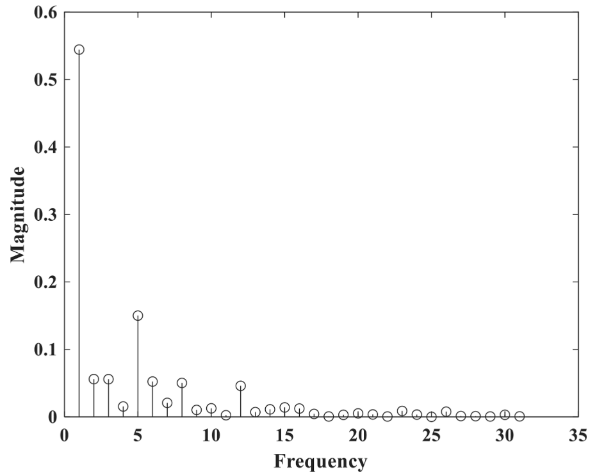

Fig. 3. Graph Fourier spectrum of the pressure signal of the Hanoi network.

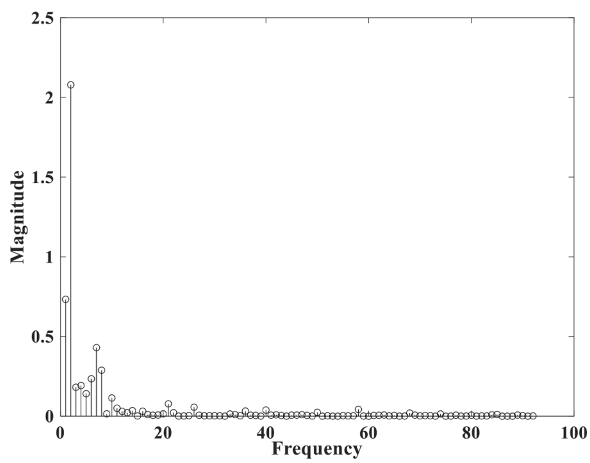

Fig. 4. Graphical Fourier spectrum of the pressure signal of Net 3.

is set to within 1% of the average daily water demand. A total of 1500 leaks were simulated for 31 nodes of the Hanoi network at a time, and random noise with a signal-to-noise ratio of 50 to 80 dB was added to the obtained pressure data to simulate environmental errors. The sensitivity matrix was obtained by the method described in Section 4.1, and the size of the sensitivity matrix was 1500 × 31. This process was repeated 6 times to obtain 6 data sets of size 1500 × 31.

The WDN used in Case 2 is the active network North Marin Water District (Net 3), and the pipe network structure is shown in Fig. 2. The system consists of 92 user nodes, 117 pipes, 2 reservoirs, 3 tanks, and 2 pumps, with a total pipeline length of 215.71 km and an average daily water demand of 3052.11 L/s. Due to the large scale of the pipe network Net 3, the workload when simulating leakage using EPANET software is large and difficult to implement, we use the WNTR toolkit in Python to simulate the pipe network. The pressure at each node during normal operation of the pipe network is first obtained, and leaks are added node by node. In the leak simulation, the size of the leak is set to 3.93% of the average daily water demand, i.e., 120 L/s for all nodes. To investigate the robustness of the algorithm under dynamics and uncertainty, we considered the variation of water demand over time, extracted pressure data in 1-h time steps, and added random noise with a signal-to-noise ratio of 50 to 80 dB to simulate environmental errors. We simulated a total of 1820 leakage events, and the size of the final generated pressure sensitivity matrix was 1820 × 92. This process was repeated 6 times to obtain 6 data sets of size 1820 × 92.

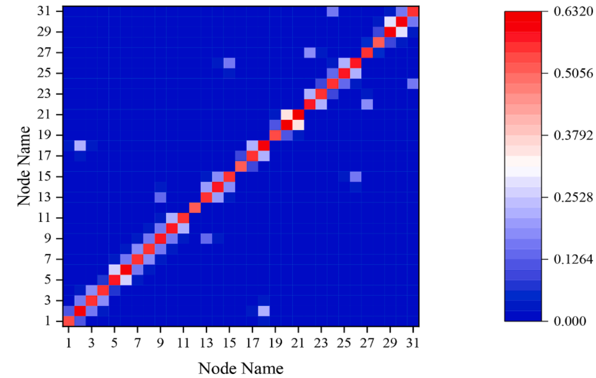

Fig. 5. Heat map of the value of the graph localization operator.

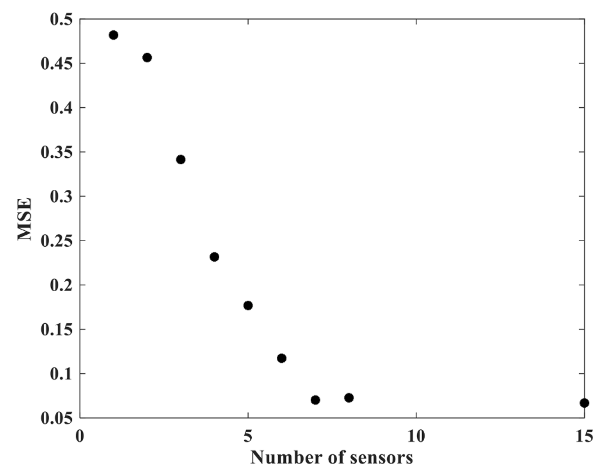

Fig. 6. Variation of the mean square error (MSE) of the reconstructed pressure signal with the increase of the number of sensors in the Hanoi network.

5.2. *Kernel function parameter setting*

The smoothing kernel function g used in both cases is the heat kernel of the negative exponential form of e (Perraudin et al., 2018), i.e.$g(\lambda) = \exp(-\tau\lambda)$. Where the value of τ is determined by the edge probability$p_s = s/N$, the current sampling ratio $p_c = |C|/N$, the normalized bandwidth $b = B/N$, and the maximum value of the eigenvalues of the graph Laplace matrix $\lambda_{\max}$, i.e., $\tau = (B|C|s)/(N^3\lambda_{\max})$. In the WDN, the water pressure at adjacent user nodes is very close to each other, which means that the pressure signals at different nodes are smooth. In other words, the pressure signal can be approximated as band-limited signal. We simulated the graph Fourier spectrum of the pressure signals of the Hanoi network and Net 3, as shown in Figs. 3 and 4, and it can be seen that the graph Fourier spectrum of the pressure signals of both networks are concentrated in the low frequency part. Based on the above analysis, we simplify the subsequent case studies by setting the bandwidth B approximation to 8 for the pressure signal when selecting the nodes in the Hanoi network and 15 for the Net 3.

5.3. *Case study 1*

In Case 1, our method uses the topology of the Hanoi network to obtain the weighted adjacency matrix of the pipe length, and the eigenvalue and eigenvector matrix are obtained by eigendecomposition using the relevant theory of graph signal processing. Thus, the graph localization operator described in Section 2.2 is constructed. Fig. 5 is the heat map that shows the value of the graph localization operator for the nodes, and it can be seen that a certain node only has a large influence on

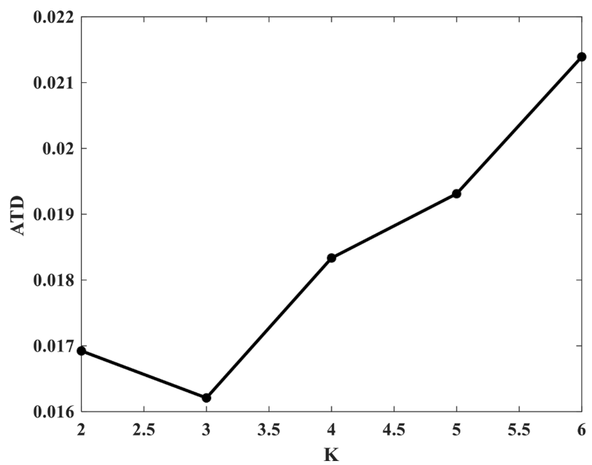

Fig. 7. Variation of the average topological distance index for leak localization with the value of K.

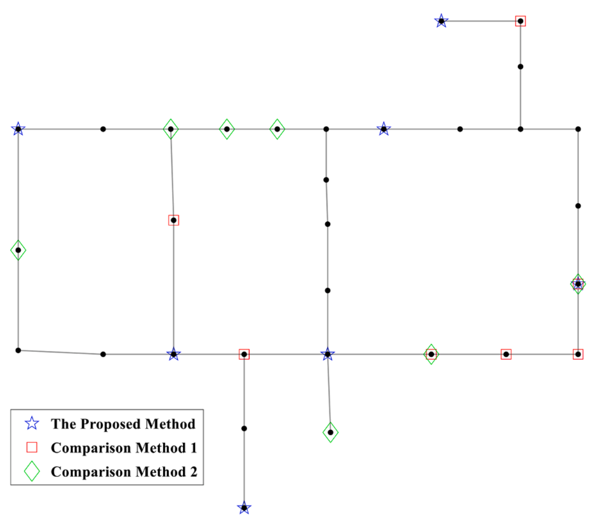

Fig. 8. Nodes selected by three different methods in the Hanoi network.

the surrounding nodes, and the larger the value the greater the influence effect. Based on the objective function, nodes that provide enough information about the graph structure and have little redundancy with the already selected nodes are selected as the locations for sensor placement.

We analyzed the effect of recovering the pressure data for all nodes with different number of sensor configurations selected according to the method introduced in Section 3.3. The mean square error (MSE) of the recovered pressure signal with the original pressure signal is shown in Fig. 6. It can be seen from the figure that as the number of sensors increases, the data recovery for the sensor nodes selected by our method becomes more and more effective. However, starting from the 7-th node, the growth rate of the recovery effect slows down. The number of sensors grows from 7 to 15 and the MSE does not change significantly. The suggested number of nodes selected using the method described in Section 3.3 is 7.

When using the KNN-based leak localization algorithm to analyze the monitoring effect of sensor nodes, different K values will have an impact on the leak localization effect. We analyzed the variation of the average topological distance index with K values when using 7 sensors for leak localization, and the changing trend is shown in Fig. 7, we choose 3 as the value of K.

We give the results of our method when 7 sensors are selected and the distribution in the network is shown in Fig. 8. The comparison method 1 for the Hanoi network is the nodes selected by Ildeberto Santos-Ruiz et al. in the paper (Santos-Ruiz et al., 2022) using information entropy theory, corresponding to nodes {3, 4, 5, 6, 11, 19, 23}. The comparison method 2 is the sensor configuration selected by Xie et al. in paper (Xie et al., 2018) using compressive sensing with sensor nodes {1, 3, 6, 24, 25, 26, 29}. The distribution in the pipe network is shown in Fig. 8 with squares and lozenges, respectively.

To obtain more accurate and reliable results, we performed leak localization validation using the 6 datasets mentioned above, and the values of evaluation metrics for our method and the two compared methods on the 6 datasets are given in Tables 3–5. The evaluation metrics used are the average topological distance index for leakage localization and the average topological distance deviation index. Table 6 shows the average values of the two leakage evaluation metrics, and the time required to select the same number of sensors for our method and the two comparison methods. From the data in Table 6, for the same number of sensors, the leak localization results for the selected nodes of our method are reduced by 63.65% for ξ_{ATD} and 46.91% for μ_{RMSE} compared to the nodes obtained using information entropy theory in the paper (Santos-Ruiz et al., 2022), and by 51.29% for ξ_{ATD} and 33.01% for μ_{RMSE} compared to the nodes used in the paper (Xie et al., 2018). This means that our method is better than the other two methods in terms of both the accuracy and stability of leak localization. In addition, we also compare the time required for node selection by different algorithms in Table 6, and it can be seen that our method takes much less time than the two methods compared.

The above analysis shows that the sensor nodes placed using our method can obtain more useful data information faster when monitoring the WDN, which can provide better help in locating subsequent leakage events.

Table 3
Results of the proposed method in Hanoi WDN.

Data set	ξ_{ATD}	μ_{RMSE}
1	0.011313	0.033586
2	0.011651	0.035202
3	0.013706	0.038429
4	0.029048	0.060424
5	0.019345	0.049090
6	0.012188	0.036572
Average	0.016209	0.042217

Table 4
Results of the information entropy-based method in Hanoi WDN.

Data set	ξ_{ATD}	μ_{RMSE}
1	0.040448	0.077276
2	0.040099	0.076368
3	0.041882	0.077344
4	0.058851	0.090361
5	0.045778	0.078591
6	0.040489	0.077220
Average	0.044591	0.079527

Table 5
Results of the compression-sensing based method in Hanoi WDN.

Data set	ξ_{ATD}	μ_{RMSE}
1	0.026561	0.053837
2	0.027027	0.055569
3	0.030960	0.060546
4	0.050283	0.082760
5	0.036576	0.067977
6	0.028244	0.057408
Average	0.033275	0.063016

Table 6
Comparison of the time required for different sensor placement methods and the effectiveness of leak localization in Hanoi network.

Method	Node indexes	Number of nodes selected	ξ_{ATD}	μ_{RMSE}	Time(s)
The Proposed	2,6,12,14,21,22,30	7	0.016209	0.042217	0.01974
Comparison method 1	3,4,5,6,11,19,23	7	0.044591	0.079527	10.150052
Comparison method 2	1,3,6,24,25,26,29	7	0.033275	0.063016	11.111954

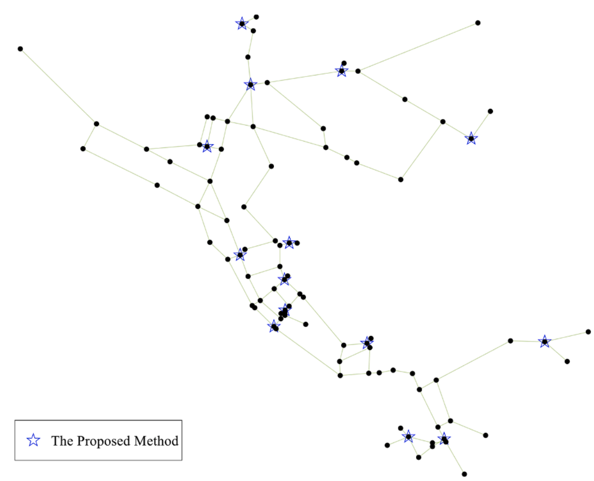

Fig. 9. Nodes selected by the proposed method in Net 3.

5.4. Case study 2

In Case 2, we compare the effectiveness of our method with the other two methods when the same number of sensors are selected. The number of sensors selected using the method in Section 3.3 is 14, corresponding to nodes {121, 127, 143, 164, 169, 181, 185, 193, 201,217, 239, 255, 261, 601}, distributed in the pipe network as shown in Fig. 9.

Following the evaluation method of the Hanoi network, the nodes obtained by our provided method are compared with the nodes selected using information entropy and compressed sensing methods. The nodes selected using information entropy are {40, 173, 177, 179, 183, 189, 199, 206, 207, 208, 267, 271, 273, 275}. The distribution in Net 3 is shown in Fig. 10(a). The nodes selected using the compressed sensing method are {10, 40, 101, 103, 105, 161, 163, 173, 193, 204, 213, 247, 267, 273}, and the distribution in Net 3 is shown in Fig. 10(b). In terms of distribution, the nodes selected using information entropy are mainly concentrated in the middle and lower parts of the pipe network, and the nodes selected using the compressive sensing method have no nodes in the upper right part of the pipe network. In contrast, the nodes selected by us are more uniform in the pipe network. Similarly, to obtain more accurate and reliable results, we performed leak localization validation using 6 datasets generated by Net 3 and give the values of evaluation metrics for our method and the two compared methods in Tables 7–9.

We collate the average values of the two leak evaluation metrics in Table 10, as well as the time required to select the nodes. Compared with the information entropy-based method, our proposed method has a ξ_{ATD} reduction of 51.14% and μ_{RMSE} reduction of 50.70%; compared to the nodes selected by the compressed sensing method, ξ_{ATD} reduction is 43.14% and μ_{RMSE} reduction is 45.19%. The comparison results show that our method achieves better leak location results even after scaling up the pipe network. Also, the time to select the nodes does not increase significantly with the increase of the network scale, compared to the other two methods which require significantly more time. Therefore,

Table 7
Results of the proposed method in Net 3.

Data set	ξ_{ATD}	μ_{RMSE}
1	0.017200	0.025210
2	0.016912	0.024857
3	0.017417	0.025518
4	0.019150	0.030038
5	0.017021	0.025159
6	0.017477	0.025560
Average	0.017530	0.026057

Table 8
Results of the information entropy-based method in Net 3.

Data set	ξ_{ATD}	μ_{RMSE}
1	0.035064	0.051705
2	0.035701	0.052726
3	0.035845	0.053126
4	0.037442	0.054555
5	0.035873	0.053534
6	0.035353	0.051449
Average	0.035880	0.052849

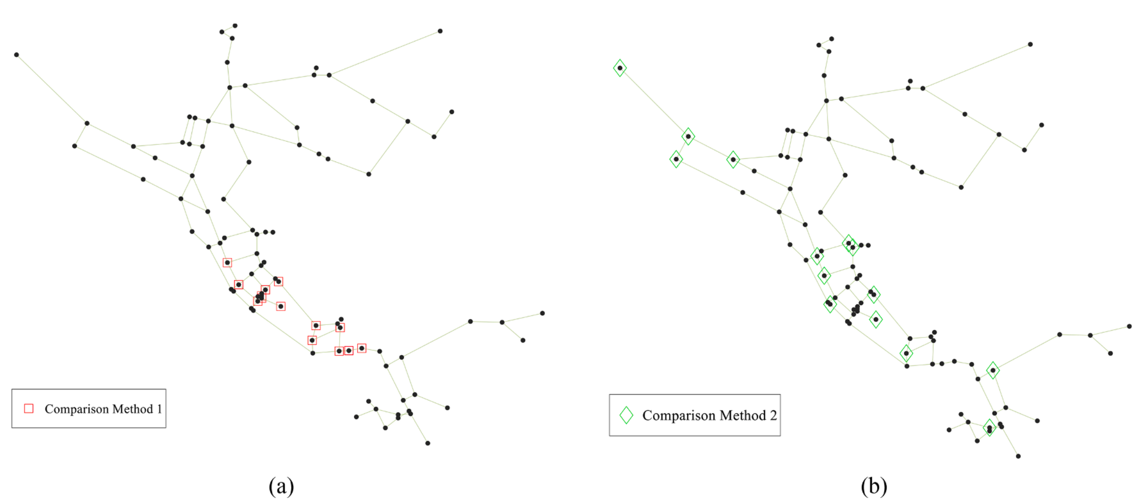

Fig. 10. Sensor nodes used in the comparison methods in Net 3. (a) Comparison method 1. (b) Comparison method 2.

Table 9
Results of the compression-sensing based method in Net 3.

Data set	ξ_{ATD}	μ_{RMSE}
1	0.032307	0.050307
2	0.031007	0.047122
3	0.031424	0.048503
4	0.028960	0.044940
5	0.031872	0.049735
6	0.029404	0.044660
Average	0.030829	0.047544

Table 10
Comparison of the time required for different sensor placement methods and the effectiveness of leak localization in Net 3.

Method	Number of nodes selected	ξ_{ATD}	μ_{RMSE}	Time(s)
The Proposed	14	0.017530	0.026057	0.003497
Comparison method 1	14	0.035880	0.052849	202.336253
Comparison method 2	14	0.030829	0.047544	15.919242

our method has better scalability to better monitor the sensor network and obtain more accurate information about the location of leaky nodes.

5.5. Scalability of the proposed method to real WDNs

The results of selecting about 5% of the sensors in the total network nodes can provide a more realistic idea about the performance of the proposed method in a real WDN. We give the results of different methods to place 2 (around 5%) and 3 (around 10%) sensors in the Hanoi case in Table 11 and 5 (around 5%) sensors in Net 3 in Table 12.

From the data in Table 11, it is easy to see that in the Hanoi case, the results of our method are not as good as the two comparison methods when we select two nodes. However, when we increase the number of selected nodes from two to three, the accuracy of leak location is significantly improved. This is when our method is better than the comparison methods. The reason for this effect is that the hydraulic behavior of the network is relatively simple in the small WDN. At this point the nodes selected based on the hydraulic model may be more accurate because it takes into account the hydraulic characteristics and flow behavior of the WDN. The nodes selected based on the sensor network structure may not accurately capture the key nodes or sensitive areas in the network. When the number of sensors increases, the sensor network structure provides a more comprehensive coverage of the entire hydraulic system, including important monitoring areas and nodes. The increased density of sensors provides more accurate and comprehensive data information, leading to a better understanding of the system's behavior. Despite the convenience of using a hydraulic model for sensor placement, it is also true that many water utilities do not have reliable hydraulic models and may not prioritize investments in periodic recalibrations. Therefore, the cost of properly placing additional nodes is acceptable compared to the cost of obtaining a reliable hydraulic model prior to placing monitoring nodes.

Moreover, in the relatively large network Net 3, our method gives better results than the comparison methods when 5 (around 5%) sensors are placed. It can be seen that the complexity of the network increases as the size of the WDN increases. In this case, the nodes selected based on the structure of the WDN are more suitable for monitoring the state of the whole network. On the contrary, nodes selected based on the hydraulic model may be limited by model simplifications or assumptions and cannot fully reflect the behavior of the real WDN.

6. Conclusion

In this study, a novel heuristic algorithm benefits from feature selection and graph signal processing theory is proposed to deploy sensors to monitor leakage events in the water distribution network and the optimal number of sensors is analyzed in conjunction with the reconstruction error of the pressure signal. The given method requires only the topological information of the WDN when selecting the sensor layout nodes, thus avoiding redundant analysis of the hydraulic simulation data of the network, which makes our method easily scalable to large networks. Then, the proposed method is applied to two different types of WDNs. In the Hanoi network, we analyzed the ability of the provided sensor placement schemes to monitor leak events of different sizes. The robustness of the algorithm is analyzed in Net 3 when the water demand of the user nodes varies with time. The results show that our method is generalizable and effective in comparison with other methods, and can significantly improve the efficiency of leak detection in WDN.

Testing on small simulated networks provides us with initial insight into the performance and effectiveness of the algorithm, but testing on large-scale real networks may face more challenges. Next, we outline how to extend these results to large-scale real networks and provide directions for future work: (1) First, we need to consider the scalability of the data scale. Large-scale real networks require us to deal with larger amounts of data at larger scales, including more nodes and connections. (2) Second, we also need to consider the dynamics of the network. Nodes and connections in real networks may change, so our algorithms need to be able to adapt to changes in the network topology. (3) Finally, we can

Table 12
Results of different methods of placing 5 (around 5%) sensors in Net 3.

Data set	ξ_{ATD}		
	The Proposed	Comparison method 1	Comparison method 2
1	0.034086	0.040890	0.041981
2	0.034200	0.040909	0.044141
3	0.036456	0.041153	0.047327
4	0.047474	0.043333	0.046301
5	0.035240	0.040604	0.045674
6	0.044593	0.039753	0.044130
Average	0.038675	0.041107	0.044926

Table 11
Results of different methods of placing 2 (around 5%) and 3 (around 10%) sensors in Hanoi network.

Data set	ξ_{ATD}					
	2 sensors			3 sensors		
	The Proposed	Comparison method 1	Comparison method 2	The Proposed	Comparison method 1	Comparison method 2
1	0.085682	0.046231	0.071021	0.016931	0.038545	0.052911
2	0.085365	0.053594	0.071726	0.017163	0.042401	0.052645
3	0.088594	0.066956	0.072184	0.025797	0.050849	0.056566
4	0.109890	0.091367	0.092987	0.047571	0.079094	0.079585
5	0.101410	0.073727	0.080698	0.035619	0.061278	0.064335
6	0.082121	0.058664	0.072239	0.020317	0.044867	0.053376
Average	0.092177	0.065090	0.076809	0.027233	0.052839	0.059903

also think about how to apply distributed computing techniques to extend our algorithm.

In our future work, we will work on solving these challenges and applying our algorithms to large-scale real networks.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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